PRE-ALGEBRA (8TH) · UNIT 5: PATTERNS AND BIVARIATE DATA

Screen Time & Sleep Study

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students analyze screen time and sleep data for the school health committee with scatter plots, a line of best fit, residuals, and two-way tables.

The project stresses honest conclusions: association is not causation, and predictions far from the data are unreliable.

Standards Addressed

- 8.SP.A.1 Scatter plots, clustering, outliers, association
- 8.SP.A.2 Lines of best fit
- 8.SP.A.3 Interpret slope and intercept
- 8.SP.A.4 Two-way tables and relative frequencies

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with the question "Does screen time affect sleep?" Part 1: scatter plot and association.
Day 2	Part 2: line of fit, predictions, and residuals.
Day 3	Part 3: two-way table and research poster.

Materials

- Student handout with the data cards
- Graph paper or a spreadsheet
- Rulers or spaghetti strands for lines of fit
- Poster paper

How the Three Versions Differ

Level	What changes
CP2	Plotting, one line of fit with predictions, and comparing two percents.
CP1	Writing a line, residuals, extrapolation, relative frequencies, and causation.
Honors	Point of averages, residual analysis, model-based limits, experiment design, and scaling to the whole school.

Watch For (Common Misconceptions)

- Saying the model proves screen time causes less sleep.
- Comparing counts instead of percents in the two-way table.
- Trusting predictions far outside the data.

Extension Ideas

- Collect anonymous class data and compare it with the study.
- Use a spreadsheet to find the least-squares line.
- Make a residual plot and describe it.

Answer Key – CP2

Part 1: Scatter Plot

#	Task	Answer
1	Make a scatter plot with screen time on the x-axis and sleep on the y-axis. Describe the association.	Negative, linear association: more screen time goes with less sleep.
2	A new student reports 8 hours of screen time and 9 hours of sleep. Is this point an outlier? Explain.	Yes. Students with about 8 hours of screen time sleep about 6 hours, so 9 hours is far from the pattern.
3	Use the line sleep ≈ −0.45 × screen + 9.4 to predict sleep for 7 hours of screen time.	−0.45(7) + 9.4 = 6.25 ≈ 6.3 hours.

Part 2: The Line of Fit

#	Task	Answer
4	Predict the sleep for 3.5 hours of screen time.	−0.45(3.5) + 9.4 = 7.825 ≈ 7.8 hours.
5	What does the slope −0.45 mean?	Each extra hour of screen time goes with about 0.45 hour (27 minutes) less sleep.
6	What does the y-intercept 9.4 mean?	The predicted sleep for a student with no screen time: about 9.4 hours.

Part 3: Two-Way Table

#	Task	Answer
7	How many students have a phone in the bedroom? How many sleep 8 or more hours?	30 have a phone in the bedroom; 6 + 14 = 20 sleep 8+ hours.
8	What percent of students with a phone in the bedroom sleep 8+ hours? What percent of students without one?	Phone: $\frac{6}{30}$ = 20%. No phone: $\frac{14}{20}$ = 70%.
9	Is there an association between a phone in the bedroom and sleeping 8+ hours? Explain.	Yes: 70% vs. 20% is a big difference.

Reflection

#	Prompt	Look for
R1	What does a negative association look like on a scatter plot?	The points go down from left to right.
R2	Why compare percents instead of counts in the two-way table?	The groups have different sizes (30 and 20), so percents make the comparison fair.

Answer Key – CP1

Part 1: Patterns

#	Task	Answer
1	Describe the direction, form, and strength of the association.	Negative, linear, and strong: the points lie close to a line that falls to the right.
2	Draw a line through (1, 9) and (8, 5.8). Find its slope and equation.	Slope = $\frac{5.8-9}{8-1}$ ≈ −0.46. y ≈ −0.46x + 9.46.
3	Compare your line with the model y = −0.45x + 9.4. Are they close? Explain.	Yes: slopes −0.46 and −0.45 and intercepts 9.46 and 9.4 give nearly the same predictions.

Part 2: Predictions & Residuals

#	Task	Answer
4	Predict sleep for 5.5 hours of screen time.	−0.45(5.5) + 9.4 = 6.925 ≈ 6.9 hours.
5	Find the residual for the student with 5 hours of screen time and 7.0 hours of sleep.	Predicted 7.15; residual = 7.0 − 7.15 = −0.15 hour (slightly less sleep than predicted).
6	Use the model to predict sleep for 20 hours of screen time. Is this prediction reasonable?	−0.45(20) + 9.4 = 0.4 hour. Not reasonable: 20 hours is far outside the data (extrapolation).

Part 3: Relative Frequencies

#	Task	Answer
7	Make a row relative-frequency table (percents within each row).	Phone: 20% sleep 8+, 80% sleep < 8. No phone: 70% sleep 8+, 30% sleep < 8.
8	What percent of all 50 students have a phone in the bedroom and sleep 8+ hours?	$\frac{6}{50}$ = 12%.
9	Does the table prove that phones in bedrooms cause less sleep? Explain.	No. It shows an association, but other factors (like bedtime rules) could explain it; an experiment is needed to show cause.

Reflection

#	Prompt	Look for
R1	What does a negative residual mean?	The actual value is below the prediction from the line.
R2	Why is extrapolation risky?	The pattern may not continue outside the range of the data.

Answer Key – Honors

Part 1: Building the Model

#	Task	Answer
1	Write the line through (2, 8.6) and (6, 6.6). Compare it with y = −0.45x + 9.4.	Slope −0.5; y = −0.5x + 9.6. It is a little steeper than the model.
2	Find the mean screen time and the mean sleep. Does the model pass close to (mean x, mean y)?	Mean x = 42.5 ÷ 10 = 4.25. Mean y = 74.9 ÷ 10 = 7.49. Model at 4.25: 7.4875 ≈ 7.49, so yes.
3	Find the residuals for x = 2 and x = 8 using the model.	x = 2: predicted 8.5, residual 0.1. x = 8: predicted 5.8, residual 0.

Part 2: Using the Model

#	Task	Answer
4	Doctors recommend at least 8 hours of sleep. According to the model, what is the most screen time that predicts 8 hours?	−0.45x + 9.4 = 8 → x ≈ 3.1 hours.
5	About how much less sleep does the model predict for a student with 6 hours of screen time than for one with 2 hours?	4 hours × 0.45 = 1.8 hours less.
6	Describe an experiment that could test whether screen time causes less sleep.	Answers vary: randomly assign volunteers to limited or normal evening screen time for two weeks and compare average sleep.

Part 3: Categorical Data

#	Task	Answer
7	Find the difference in the percent who sleep 8+ hours between the two groups.	70% − 20% = 50 percentage points.
8	If the whole school of 600 students had the same pattern, how many would have a phone in the bedroom and sleep 8+ hours?	12% of 600 = 72 students.
9	Write your own survey question pair for a two-way table about sleep, and explain what result would show an association.	Answers vary; an association is shown when the row percents differ noticeably between groups.

Reflection

#	Prompt	Look for
R1	Why does a good line of fit pass near the point of averages?	Balancing the points above and below the line keeps it near the center of the data, which is the mean point.

#	Prompt	Look for
R2	How is a residual plot useful?	Residuals scattered randomly around 0 suggest a linear model fits; a pattern suggests a curve fits better.